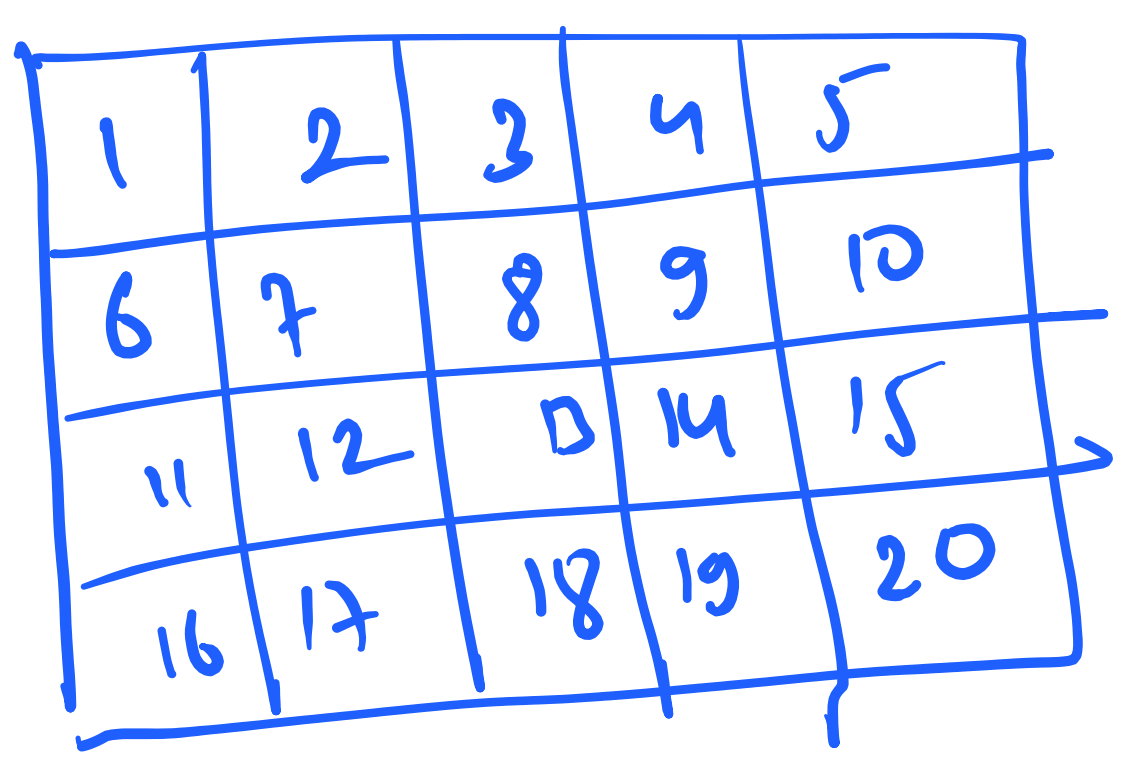
Lec-17

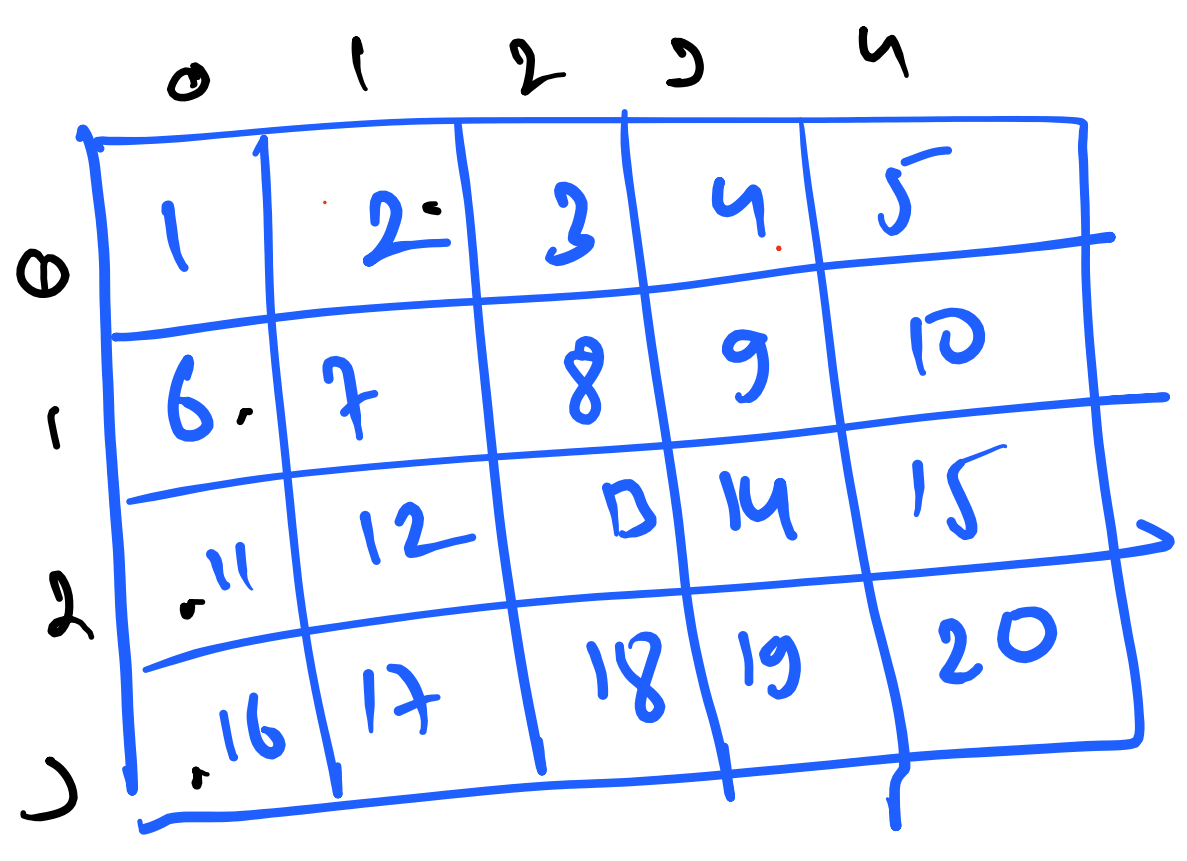
02/11/26 2:26 PM



1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

1 6 11 16 17 12 7 2 3 8 13 18 19 14 9 4

5 10 15 20



	0	1	2	3	4
0	1	2	3	4	5
1	6	7	8	9	10
2	11	12	13	14	15
3	16	17	18	19	20

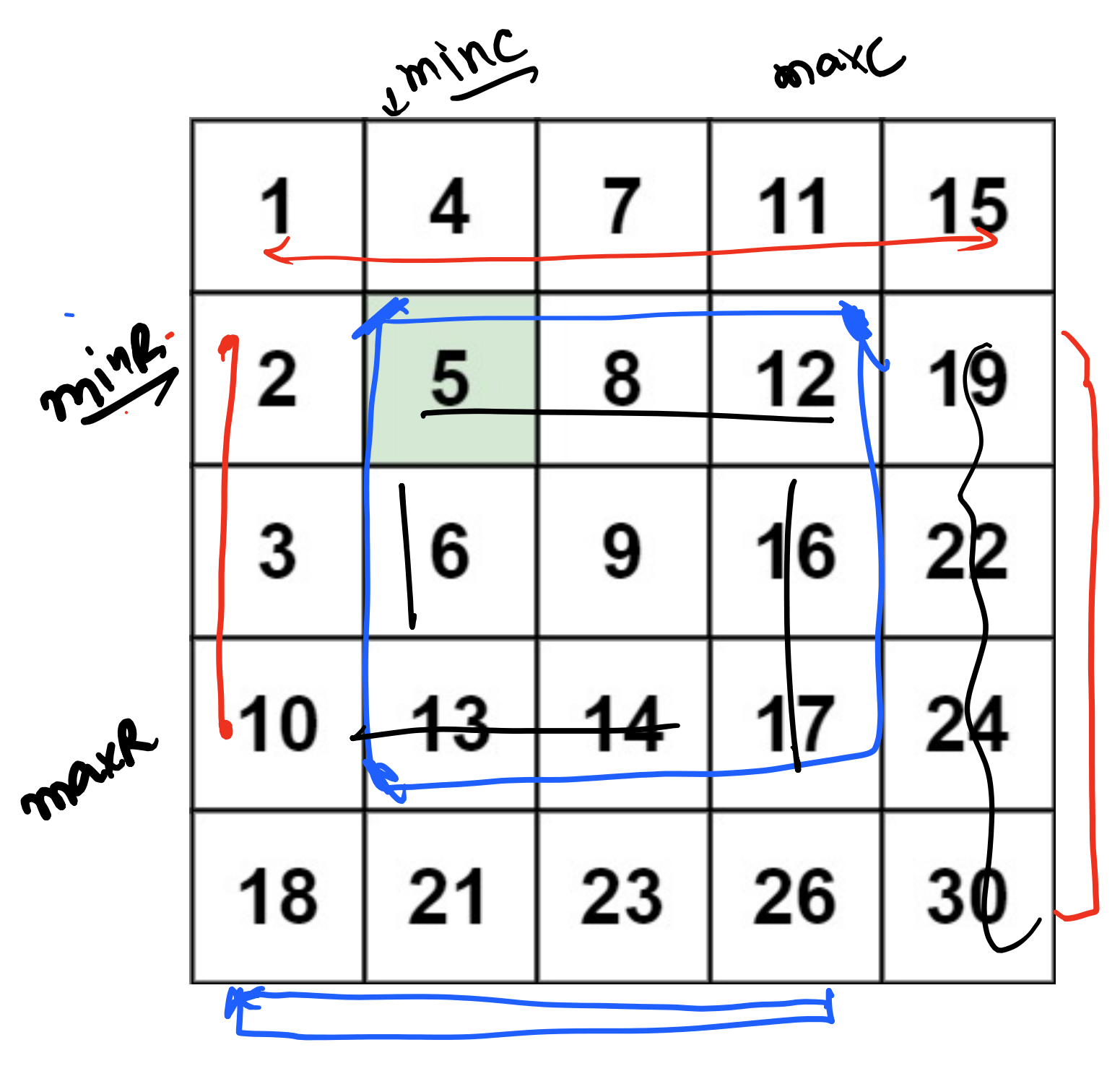
i j

0-0	3-1	0-2	3-3	0-4
1-0	2-1	1-2	2-3	1-4
2-0	1-1	2-2	1-3	2-4
3-0	0-1	3-2	0-3	0-4

```
for( col=0   col< arr[0].len   col++){
    if( col%2==0){
    for( row=0   row< arr.len   row++)

    }
    else{
    for( row= arr.len-1   row>=0 row--)

    }
}
```



1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	~~13~~	~~14~~	17	24
18	21	23	26	30

1 4 7 11 15 19 22 24 30 26 23 21 8 10

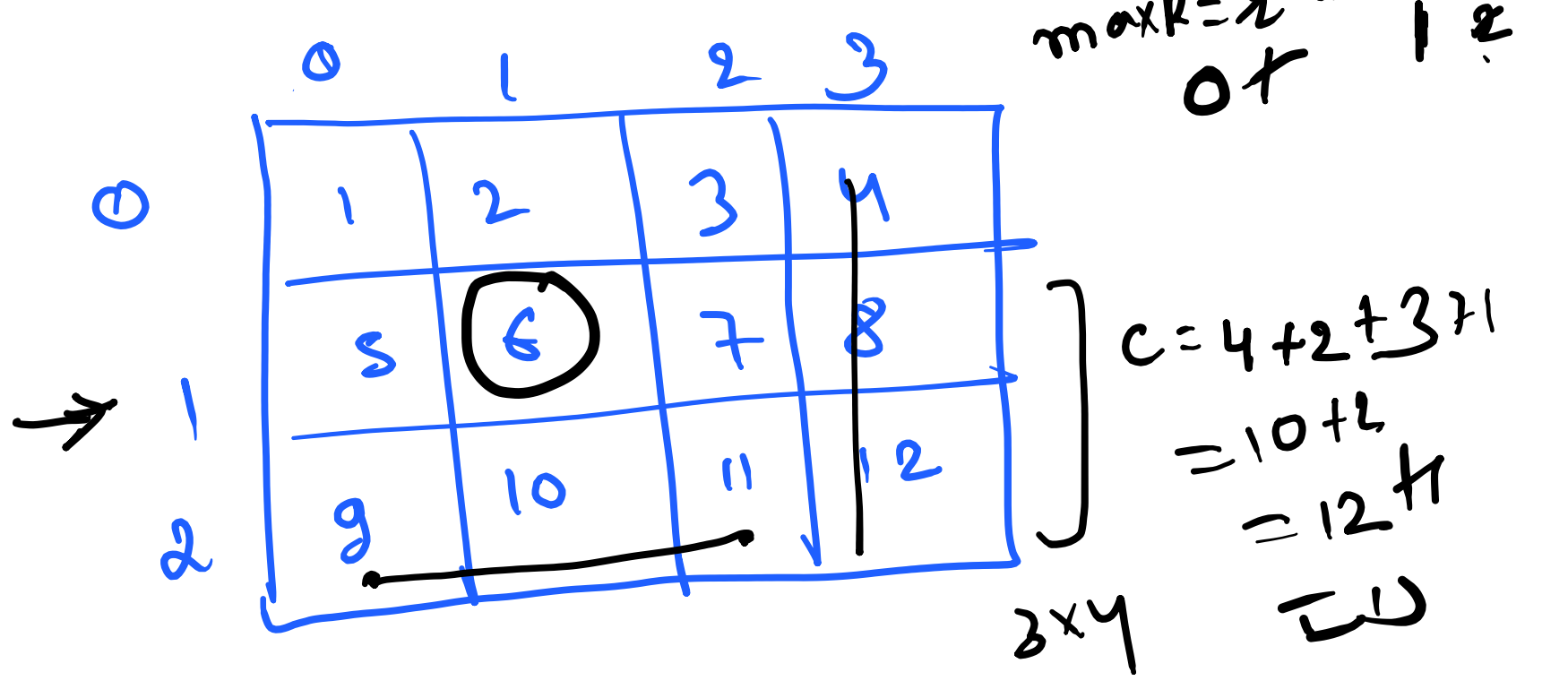
3 2 5 8 12 16 17 14 13 6 9

```
{ for(i= minc   i<= maxc   i++){
    syso (arr[minR][i])
  }
  minR++
  for(i=minR   i<=maxR i++){
    syso (arr[i][maxc])
  }
  maxc--
  for(i= maxc   i>=minc   i--){
    syso (arr[maxR][i])
  }
  maxR--
  for(i=maxR   i>=minR   i--){
    syso (arr[i](minc)
  }
  minc++
```

```
public static void Print(int[][] arr) {
      int minr = 0, minc = 0, maxr = arr.length - 1, maxc = arr[0].length - 1;
      int te = arr.length * arr[0].length; ->12
    //int c = 0;
      while (c < te) { 0<12 11<12 D
         for (int i = minc; i <= maxc; i++) {
            System.out.print(arr[minr][i] + " ");
            c++;
         }
         minr++;
         for (int i = minr; i <= maxr; i++) {
            System.out.print(arr[i][maxc] + " ");
            c++;
         }
         maxc--;
         for (int i = maxc; i >= minc; i--) {
            System.out.print(arr[maxr][i] + " ");
            c++;
         }
         maxr--;
         for (int i = maxr; i >= minr; i--) {
            System.out.print(arr[i][minc] + " ");
            c++;
         }
         minc++;
      }

   }
```

minR=~~0~~ 1, minc=~~0~~ 1
maxR=2 1, maxc=~~3~~ 2
0 1 2

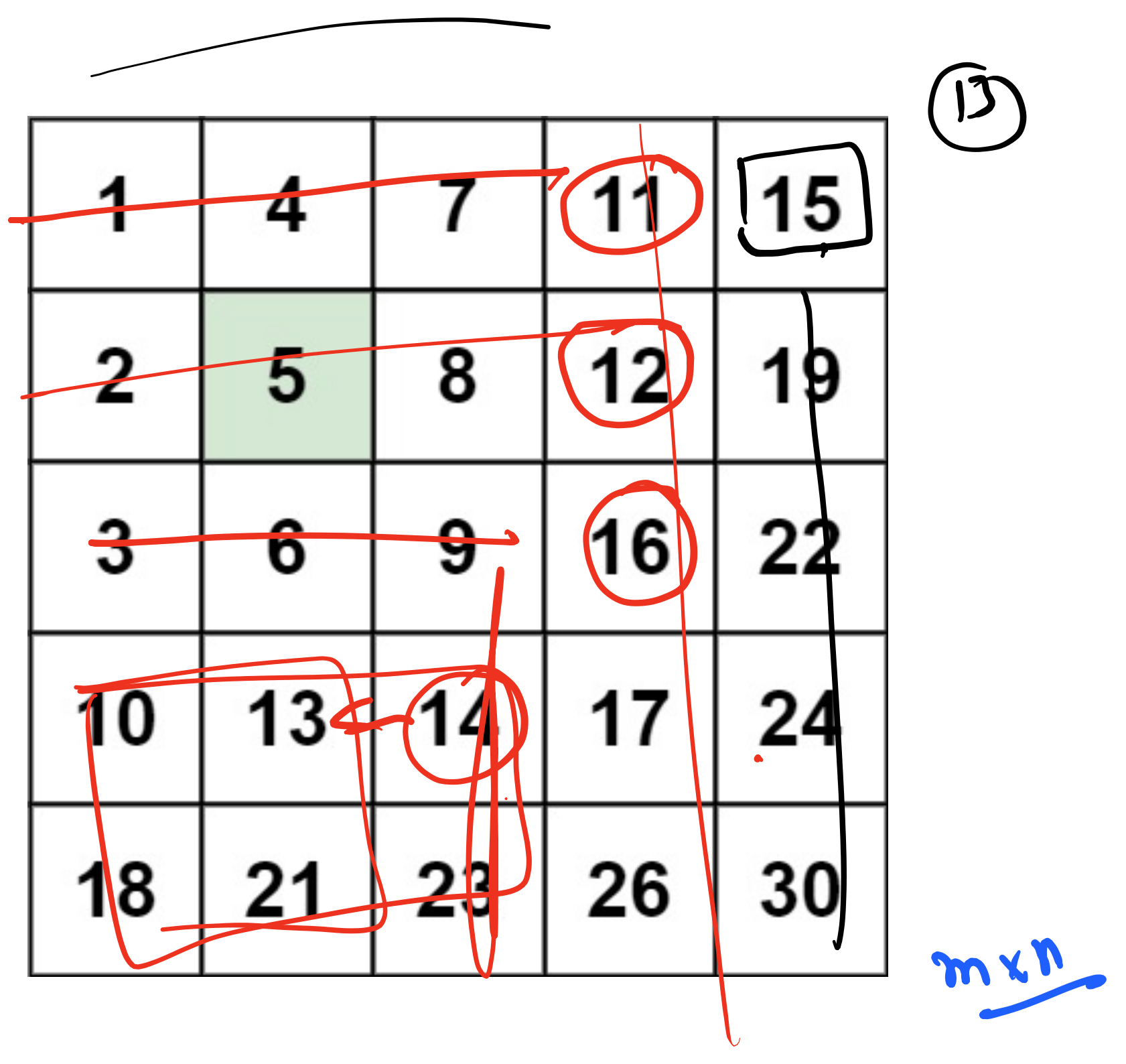


	0	1	2	3
0	1	2	3	4
1	5	6	7	8
2	9	10	11	12

c=4+2+3+1
=10+2
=12+1

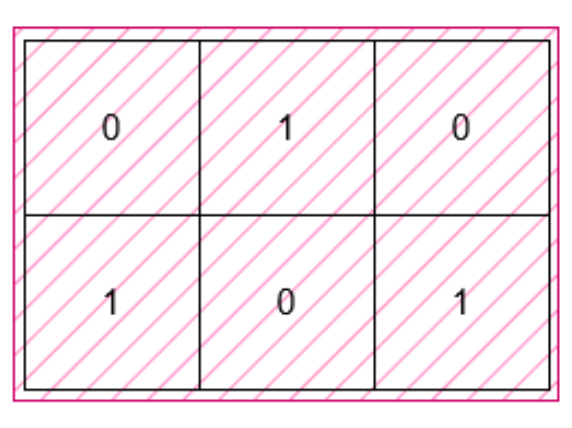
3x4

1 2 3 4 8 12 11 10 9 5 6 7 6

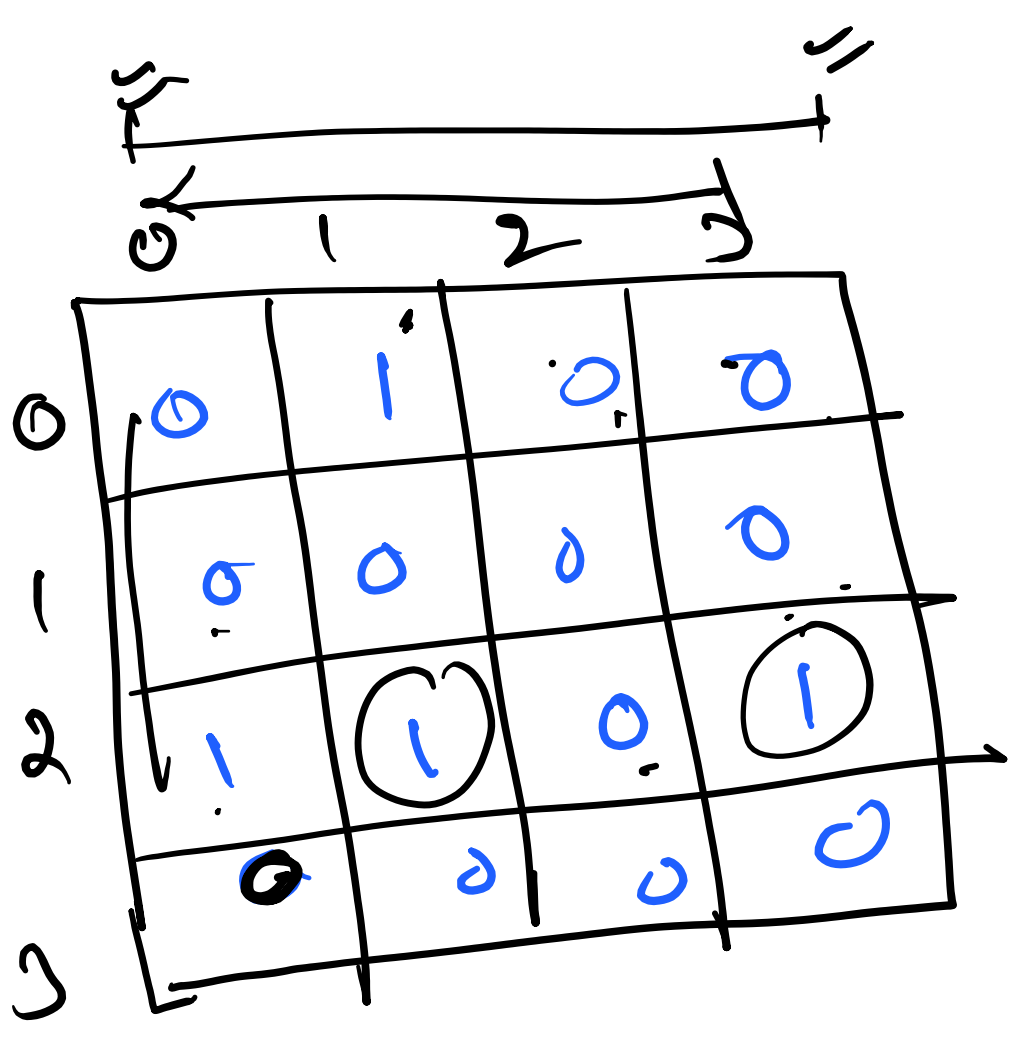


1	4	7	11	15
2	5	8	12	19
3	6	9	16	22
10	13	14	17	24
18	21	23	26	30

```
Row=0,  col=arr[0].len-1
while (row< arr.len  && col>=0){
 if (arr[row][col]==target) {
     Return
 }
 else if (arr[row][col]>target){
     col--
 }
 else{
     row++
 }
 }
}
```



0	1	0
1	0	1



	0	1	2	3
0	0	1	0	0
1	0	0	0	0
2	1	1	0	1
3	0	0	0	0

2,0
1,1

2,1

2,1)

minR=0
minc=0

maxR=2
maxc=~~1~~ 3

3 ← 0 → 2

4 ← 0 → 3

```
for (int i = 0; i < grid.length; i++) {
        for (int j = 0; j < grid[0].length; j++) {
           if (grid[i][j] == 1) {
              minr = Math.min(minr, i);
              minc = Math.min(minc, j);
              maxr = Math.max(maxr, i);
              maxc = Math.max(maxc, j);
           }
        }

     }
```

maxR-min+1 — l

maxc-min+1 — b